Data cleaning

Q1. What is data cleaning, and why is it important in data analysis?

- What are the potential consequences of analyzing unclean or messy data?
- Explain the common steps involved in cleaning and organizing data.

Ans: Data cleaning is the process of identifying and fixing problems in a dataset so that it is accurate, consistent, and ready for analysis.it is important to get accurate output or result basis on that we could take correct decision.

Analyzing **unclean or messy data** can lead to serious problems at every stage of data analysis. like Incorrect or Misleading Results, Poor Decision-Making & Biased Analysis and Models, Reduced Performance, Increased Costs and Time Waste

Below are the common steps involved in cleaning organizing data.

1. Data Inspection and Understanding
2. Handling Missing Data
3. Removing Duplicates
4. Correcting Errors and Inconsistencies
5. Standardizing Formats
6. Managing Outliers
7. Data Transformation
8. Data Validation and Quality Checks
9. Data Organization and Documentation

Q2. How would you sort the following dataset first by "Department" (A-Z) and then by "Salary" (Largest to Smallest)? Write a step-by-step approach?

Ans: Details steps of sorting as per below.

1. **Open the dataset in Excel**
    - Ensure the data is in a table format with column headers (e.g., *Name*, *Department*, *Salary*).
2. **Select the entire dataset**
    - Click any cell inside the dataset.
    - Press **Ctrl + A** to select all data.
    - Make sure the headers are included.

3. **Go to the Data tab**
    - On the Excel ribbon, click the **Data** tab.
4. **Open the Sort dialog box**
    - Click the **Sort** button.
    - The Sort dialog box will appear.
5. **Set the primary sort (Department)**
    - In the **Sort by** dropdown, select **Department**.
    - Set **Sort On** to **Cell Values**.
    - Set **Order** to **A to Z**.
6. **Add a secondary sort level (Salary)**
    - Click **Add Level**.
    - In the new row:
        - **Then by**: select **Salary**
        - **Sort On**: Cell Values
        - **Order**: **Largest to Smallest**
7. **Confirm headers**
    - Ensure the checkbox **"My data has headers"** is checked.
8. **Apply the sort**
    - Click **OK**.
9. **Verify the results**
    - Confirm departments are listed alphabetically.
    - Check that salaries decrease from highest to lowest within each department.

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Q3. Explain the use of text functions such as TRIM , LEFT, RIGHT, MID, and CONCAT in data cleaning?

Ans: Below is an explanation of each function and how it helps clean data:

1. **TRIM:** Removes extra spaces from text. Eliminates leading, trailing, and multiple internal spaces. Fixes inconsistencies that cause mismatches in text values.

Example: =TRIM(A2) Before: " Department " After: "Department"

2. **LEFT:** Extracts a specified number of characters from the beginning of a text string. Pulls codes, prefixes, or country identifiers from longer text. Helps standardize identifiers.

Example: = LEFT (A2, 3) Extracts the first 3 characters from a value like "USA-1023" → "USA"

3. **RIGHT:** Extracts a specified number of characters from the end of a text string. Retrieves suffixes, IDs, or year values. Removes unwanted leading text.

Example: =RIGHT (A2, 4) From "Excel_2025" → "2025"

4. **MID:** Extracts text from the middle of a string based on a starting position and length. Pulls embedded values such as area codes, product numbers, or dates. Helps split complex text fields.

Example: =MID (A2, 5, 2) From "EMP-45-NY" → "45"

5. **CONCAT:** Combines text from multiple cells or text strings into one.

- Merges separated text fields (e.g., first name and last name).
- Reconstructs cleaned or standardized values.

Example: =CONCAT (A2, " ", B2) Combines "Satyam" and "Dubey" → "Satyam Dubey"

Q4. What is the role of date functions like TODAY in managing datasets?

Ans: Date functions like TODAY play an important role in managing and analyzing datasets by making data dynamic, timely, and easier to maintain. Eliminates the need to manually enter the current date repeatedly. Keeps datasets current and relevant.

Below are the key roles

1. Automatically Track Current Dates

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2. Enable Dynamic Calculations
3. Support Time-Based Analysis
4. Improve Data Validation and Accuracy
5. Assist in Automation and Reporting
6. Help with Conditional Formatting
7. Simplify Data Management

Q5. Apply Data Validation to restrict Quantity values to only whole numbers between 1 and 10. Configure an input message that appears when a user selects a cell in the "Quantity" column explaining: "Please enter a whole number between 1 and 10."

Set up an error alert message that triggers if the user enters a number less than 1 or greater than 10, showing: "Invalid input! The quantity must be a whole number between 1 and 10."

Write a step-by-step approach for this question

Ans: **Step 1: Select the Quantity cells**

1. Open the Excel worksheet containing the table.
2. Click on the first Quantity cell (for example, **D2**).
3. Drag down to select all Quantity cells (e.g., **D2:D10**).

Step 2: Open Data Validation

1. Go to the **Data** tab on the Excel ribbon.
2. Click **Data Validation** in the *Data Tools* group.
3. The **Data Validation** dialog box will open.

Step 3: Set the validation rule (whole numbers only)

1. In the **Settings** tab:
    - **Allow:** Select **Whole number**
    - **Data:** Select **between**
    - **Minimum:** Enter 1
    - **Maximum:** Enter 10

This restricts entries to whole numbers from 1 to 10 only.

Step 4: Configure the Input Message

1. Click the **Input Message** tab.
2. Check **"Show input message when cell is selected."**
3. Enter:
    - **Title:** Quantity Instruction
    - **Input Message:**
    - Please enter a whole number between 1 and 10.

This message appears when a user clicks a Quantity cell.

Step 5: Set up the Error Alert

1. Click the **Error Alert** tab.
2. Ensure **"Show error alert after invalid data is entered"** is checked.
3. Set:
    - **Style:** Stop
    - **Title:** Invalid Input
    - **Error Message:**
    - Invalid input! The quantity must be a whole number between 1 and 10.

This message appears if a user enters a value less than 1, greater than 10, or a non-whole number.

Step 6: Apply the validation

1. Click **OK** to apply the Data Validation settings.

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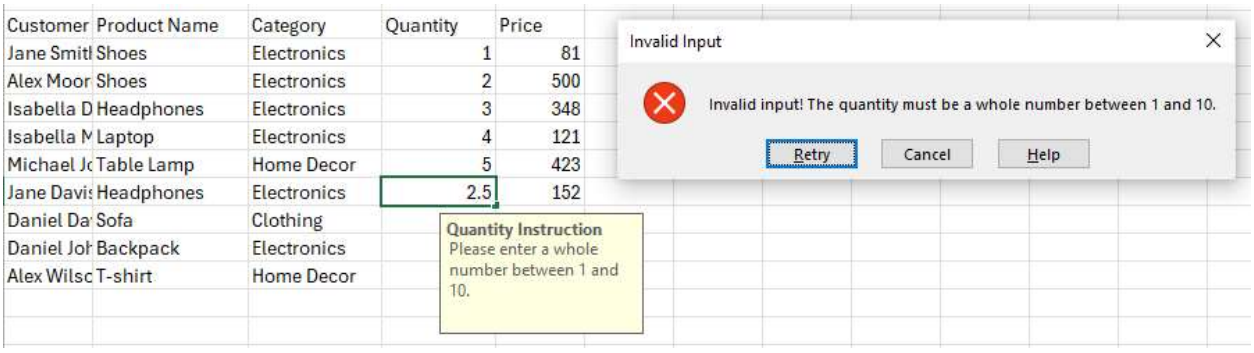
Step 7: Test the validation

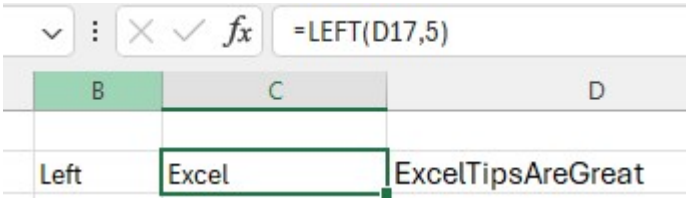
- Enter 5 → Accepted
- Enter 0, 11, 2.5, or text → Error message appears

Screen shot for reference where actual run the steps

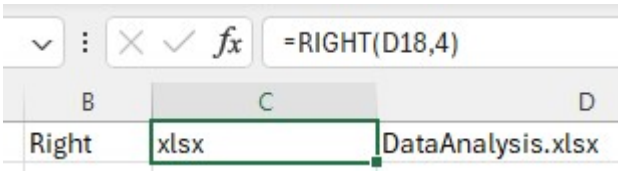


Q6. understand and apply fundamental text functions like LEFT, RIGHT, MID, and LEN.

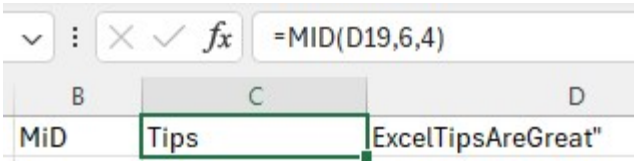
Extract the first 5 characters from the string "ExcelTipsAreGreat" using the LEFT function.



Extract the last 4 characters from "DataAnalysis.xlsx" using the RIGHT function.



Extract the substring "Tips" from "ExcelTipsAreGreat" using the MID function.



Count the total number of characters in the string "Hello World!" using the LEN function.

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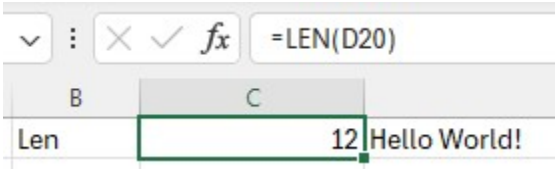
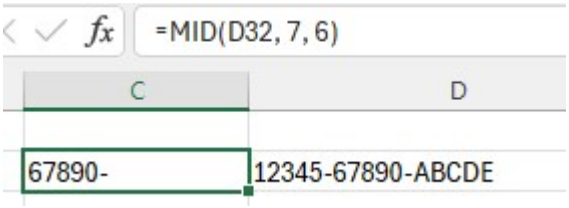
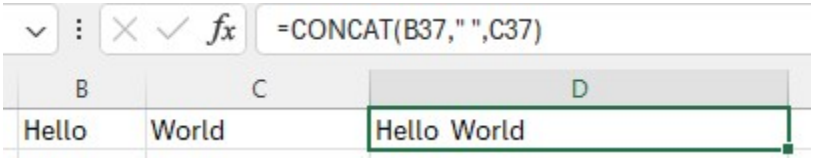


Create a formula to extract the middle 6 characters from "12345-67890-ABCDE".

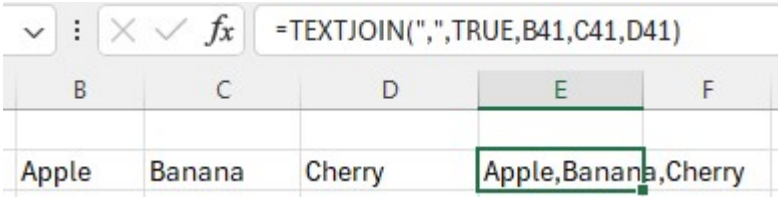


Q.7 Understand how to combine text using CONCAT, TEXTJOIN, and the & operator.

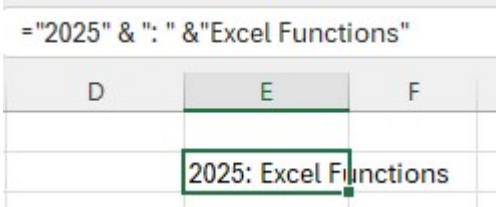
Use CONCAT to combine "Hello" and "World" with a space in between.



Combine "Apple", "Banana", and "Cherry" into a single string separated by commas using TEXTJOIN.



Use the & operator to create the string "2025: Excel Functions" by combining "2025", ": ", and "Excel Functions".



Create a comma-separated list from the range A1:A5 using TEXTJOIN

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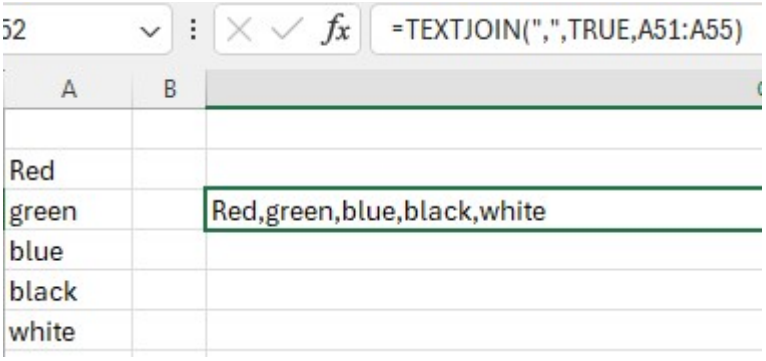
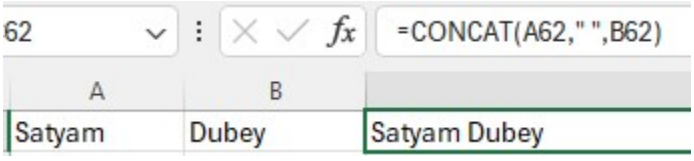


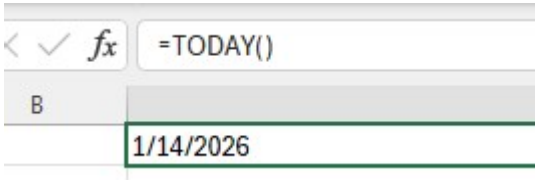
Combine first names in column A with last names in column B to create full names in column C.



Q8. Understanding TODAY() and NOW()

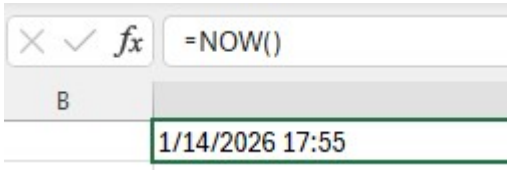
a. What is the difference between TODAY() and NOW() in Excel? Provide an example of when you would use each function.

Ans: Use **TODAY()** when you only need the current date and not the time. To automatically show today's date on a report:



Use **NOW()** when you need both the current date **and** time.

E.g To record the exact time a task was completed:



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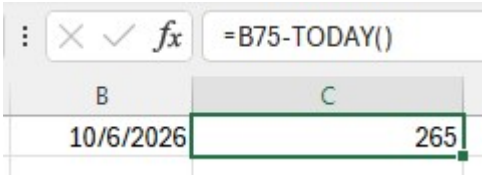
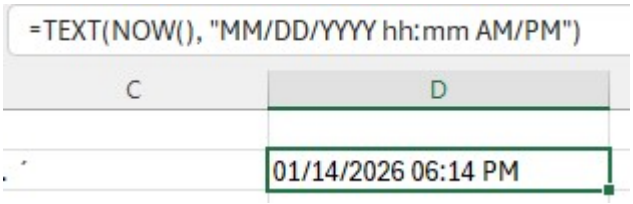
b. If cell A1 contains the date 2025-06-10, write a formula using TODAY() to determine how many days are left until that date



c. Write an Excel formula using NOW() to display the current date and time in the format MM/DD/YYYYHH:MM AM/PM. ´



d. If a cell contains =TODAY (), what will happen when the worksheet is reopened the next day? Explain
Ans: If a cell contains =TODAY (), then when the worksheet is reopened the next day, the value in that cell will automatically update to the new current date.

e. You want to store a static date (today's date) in a cell without it changing every day. What keyboard shortcut should you use?
Ans: Ctrl + ;